

Daily Content Intel

August 3, 2026

Topic A — Reel

TELEPROMPTER COPY (paste into Instagram)

Okay, it's the first week of August, which means two a days, and a whole lot of athletes are quietly eating way less than they're burning.

Here's what happens when you're in that hole. Researchers at Arkansas and the Army put 19 trained young men through 5 day energy deficits at about 30 percent below what they actually needed, kept dietary protein high the entire time, and then traced where the amino acids went. Muscle is last in line. Your liver, your immune system, your gut lining, your connective tissue, they all get paid first, and muscle gets a thank you note.

The dosing piece is the part I'd actually change on Monday. About 0.25 to 0.30 grams of protein per kilogram per feeding maxes out the muscle building signal, and when they tripled that dose, muscle protein synthesis did not budge. It did improve whole body protein balance, which is recovery, immune function, connective tissue repair, kind of everything that isn't your biceps.

So the fix is boring. Eat enough total food during camp. Then put 25 to 35 grams of protein at 4 points in the day, and make one of them right after practice.

You can't supplement your way out of a calorie hole. Does that make sense?

Be Good. Help Someone. Learn Lots.

Hook: If your athlete is dropping weight in August camp, muscle is the first thing to go.

Spoken script (word for word):

Okay, it's the first week of August, which means two a days, and a whole lot of athletes are quietly eating way less than they're burning.

Here's what happens when you're in that hole. Researchers at Arkansas and the Army put 19 trained young men through 5 day energy deficits at about 30 percent below what they actually needed, kept dietary protein high the entire time, and then traced where the amino acids went. Muscle is last in line. Your liver, your immune system, your gut lining, your connective tissue, they all get paid first, and muscle gets a thank you note.

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protein at 4 points in the day, and make one of them right after practice.

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On-screen text seeds:

- 0:00 Dropping weight at camp? Muscle goes first.
- 0:10 19 men, 5 days, 30% below need
- 0:22 Muscle is last in line for amino acids
- 0:38 0.25 to 0.30 g protein per kg, per feeding
- 0:52 They tripled the dose. Muscle signal did not move.
- 1:05 25 to 35 g protein, 4 times a day
- 1:15 Eat enough first. Then supplement.

Caption (copy-paste ready):

August camp is where a lot of athletes quietly slide into a calorie hole.

Researchers ran 19 trained young men through 5 day energy deficits at about 30% below their needs, kept dietary protein high,

and traced where the amino acids actually went. Muscle sits at the back of the line. Liver, gut, immune tissue, and wound repair get served first.

The dosing part is what I'd change tomorrow. Roughly 0.25 to 0.30 g of protein per kg per feeding maxes out the muscle signal. Tripling that dose did nothing extra for muscle protein synthesis, though it did improve whole body protein balance, which is recovery and tissue repair.

So for a high school or college athlete in two a days: eat enough total food, then put 25 to 35 g of protein at 4 points in the day, one of them right after practice.

You can't supplement your way out of a calorie hole.

Be Good. Help Someone. Learn Lots.

#footballtraining #lacrosse

#highschoolfootball #sportsnutrition

#athletedevelopment

#strengthandconditioning

#sportsphysicaltherapy

#fuelingforperformance #twoadays

#returntosport

Personalize this

- Which of your HS football guys visibly dropped weight in the first week of camp last August, and what did their actual eating day look like once you asked them to walk you through it?
- Do you coach a lineman differently than a skill player on post-practice fueling during a two a day block, and does your protein target per feeding shift with body size?
- Have you ever slowed a return-to-sport progression because the athlete wasn't eating enough to repair tissue, and how did you explain that call to the athlete and the parents?

Screenshots:

- title | <https://pubmed.ncbi.nlm.nih.gov/32768315/>
| "Effects of high versus standard essential amino acid intakes on whole-body protein turnover and mixed muscle protein synthesis during energy deficit: A randomized, crossover study"

- physiology | <https://pubmed.ncbi.nlm.nih.gov/32768315/> | "Consuming 0.10-0.14 g essential amino acids (EAA)/kg/dose (0.25-0.30 g protein/kg/dose) maximally stimulates muscle protein synthesis (MPS) during energy balance."
- actionable | <https://pubmed.ncbi.nlm.nih.gov/32768315/> | "These data demonstrate that high compared to standard EAA ingestion enhances whole-body protein status during underfeeding."

Sources:

- <https://pubmed.ncbi.nlm.nih.gov/32768315/>
- <https://pubmed.ncbi.nlm.nih.gov/33413462/>

Topic B — Reel

TELEPROMPTER COPY (paste into Instagram)

Okay, hot take. Zone 2 got sold to your athletes by longevity podcasts, and now I've got football and lacrosse kids jogging 45 minutes at conversational pace in August and calling it building a base.

A 2025 narrative review in Sports Medicine went at this directly. The authors asked whether Zone 2 really is the best intensity for mitochondrial and fat oxidation capacity, and they concluded the current evidence doesn't support it, and that prioritizing intensities above Zone 2 matters more, especially when your training volume is low.

Your athlete's conditioning volume is low. They've got school, practice, lifting, and a game on Friday. A professional endurance athlete has 25 hours a week

to spend on this. Your kid has about 3.

And in youth athletes specifically, a 2025 meta analysis found high intensity intervals beat general training for VO2 max, with less total training time on the clock.

Here's what I'd rather see. 2 conditioning days a week, short repeats at the speeds the sport actually asks for, full recovery in between, and let the practice sessions carry the easy aerobic work.

The base gets built by playing the sport, right?

Be Good. Help Someone. Learn Lots.

Hook: Zone 2 is the most overrated thing in high school conditioning right now.

Spoken script (word for word):

Okay, hot take. Zone 2 got sold to your athletes by longevity podcasts, and now I've

got football and lacrosse kids jogging 45 minutes at conversational pace in August and calling it building a base.

A 2025 narrative review in Sports Medicine went at this directly. The authors asked whether Zone 2 really is the best intensity for mitochondrial and fat oxidation capacity, and they concluded the current evidence doesn't support it, and that prioritizing intensities above Zone 2 matters more, especially when your training volume is low.

Your athlete's conditioning volume is low. They've got school, practice, lifting, and a game on Friday. A professional endurance athlete has 25 hours a week to spend on this. Your kid has about 3.

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Here's what I'd rather see. 2 conditioning days a week, short repeats at the speeds the sport actually asks for, full recovery in between, and let the practice sessions carry the easy aerobic work.

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On-screen text seeds:

- 0:00 Zone 2 is overrated for your athlete.
- 0:12 45 minute jogs in August are not base building
- 0:25 Sports Medicine, 2025: evidence does not back Zone 2 as optimal
- 0:42 Pro endurance athlete: 25 hrs/week. Your kid: 3.
- 0:55 Youth meta analysis: intervals beat general training for VO2 max
- 1:08 2 conditioning days. Short repeats. Full recovery.
- 1:18 Practice is the aerobic work.

Caption (copy-paste ready):

Zone 2 got sold to high school athletes by longevity podcasts, and now I've got football and lacrosse kids jogging 45 minutes at conversational pace in August calling it base building.

A 2025 narrative review in Sports Medicine looked at whether Zone 2 is actually the best intensity for mitochondrial and fat oxidation capacity. The authors concluded the evidence doesn't support it, and that prioritizing intensities above Zone 2 matters more when training volume is low.

Your athlete's conditioning volume is low. School, practice, lifting, and a game on Friday. A professional endurance athlete has 25 hours a week to spend on this. Your kid has about 3.

In youth athletes specifically, a 2025 meta analysis found intervals beat general training for VO2 max with less total training time.

What I'd rather see: 2 conditioning days a week, short repeats at the speeds the sport actually asks for, full recovery between them, and let practice carry the easy aerobic work.

The base gets built by playing the sport.

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#footballtraining #lacrosse

#highschoolfootball #speedtraining

#conditioning #strengthandconditioning

#athletedevelopment #zone2

#sportsphysicaltherapy #offseasontraining

Personalize this

- What does your actual in-season conditioning look like for a HS football athlete on a Friday game schedule, and how many hard days do you allow in a camp week?
- Which position groups on your teams show up to camp having jogged all summer, and what does that cost them in the first week of change of direction and contact work?
- Have you had a return-to-sport athlete whose "cardio" was all steady state, and what broke down first when you put them back into repeat sprint work under fatigue?

Screenshots:

- title | <https://pubmed.ncbi.nlm.nih.gov/40560504/>
| "Much Ado About Zone 2: A Narrative Review Assessing the Efficacy of Zone 2 Training for Improving Mitochondrial

Capacity and Cardiorespiratory Fitness in the General Population"

- physiology | <https://pubmed.ncbi.nlm.nih.gov/40560504/> | "We conclude that current evidence does not support Zone 2 training as the optimal intensity for improving mitochondrial or fatty acid oxidative capacity."
- actionable | <https://pubmed.ncbi.nlm.nih.gov/40560504/> | "Further, evidence suggests prioritizing higher exercise intensities (> Zone 2) is critical to maximize cardiometabolic health benefits, particularly in the context of lower training volumes."

Sources:

- <https://pubmed.ncbi.nlm.nih.gov/40560504/>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC12497894/>

Reference

Runner-up topics

- EAAs and cognitive performance under acute stress (Ferrando, ISSN 2026 conference pilot): reaction time held near baseline with EAAs while placebo slipped roughly 28 ms. Unpublished, 8 subjects, blood draws on 5. Worth revisiting only if it gets published, and the adolescent-athlete "brain under fatigue late in a game" angle is strong.
- Adding L-tyrosine on top of a complete EAA blend gave no advantage over EAAs alone in the same pilot. Good ammo against stacked "focus" formulas marketed to athletes, once published.
- HIIT vs sport-specific drills for repeated sprint ability in male team-sport athletes (PLOS One meta-analysis): no difference for RSA, which is a useful counterweight to today's Draft B and would make a fair, self-critiquing follow-up.

Sources scanned today

- Newsletter digest:
newsletter_digest_2026-08-03.pdf (Mike T. Nelson, Weekly Wrap Up Aug 2). Two usable threads: "Muscle Is Last in Line When You Diet" (EAA dosing during energy deficit, the Arkansas/USARIEM tracer work) and "Why Zone 2 Cardio Is Stupid for Most Lifters." Also flagged: EAA + cognitive performance under stress (unpublished 8-subject conference pilot, skipped as too thin to cite).
- PubMed / Sports Medicine: Storoschuk et al. 2025 narrative review directly challenging Zone 2 as the optimal intensity. Clean, provocative, peer-reviewed.
- PubMed / Clinical Nutrition: Gwin et al. 2021 randomized double-blind crossover on standard vs high EAA during a 5-day 30% energy deficit. Gives the dose ceiling with real tracer numbers.
- Physiological Reports: Matzka et al. 2025 meta-analysis, HIIT vs alternative modalities in youth athletes. Used as the youth-specific support for Draft B.

- JISSN: Gwin et al. 2021 EAA-enriched whey vs whey vs mixed meal during energy deficit. Used as Draft A support.
- Searched but nothing new enough to lead with: BJSM / JOSPT July 2026 ACL and hamstring return-to-sport (all results were prior-year papers already covered), HS football in-season sprint programming (only coaching-blog content, no new trial).
- Adjacency note: the Research-Inbox already holds adolescent-athlete-underfueling-low-energy-availability (2026-07-15) and low-volume-hiit-athlete-engine (2026-07-08). Both drafts today come from different sources and make different claims (dose ceiling; Zone 2 specifically), so they stand on their own.